The nim-factorial of $3 \cdot 2^{n-1} - 1$ is 2: a proof of Layman's second conjecture on OEIS A059970

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Abstract

Let $a(n) = 1 \otimes 2 \otimes \cdots \otimes n$ be the nim-factorial (OEIS `A059970`). In March 2001 J. W. Layman conjectured that $a(2^n + 2^{n-1} - 1) = 2$ for all $n \geq 1$, reporting verification for $n \leq 15$. We prove it. The proof rests on two observations. First, the subspace polynomial of $\{0, \ldots, 2^t - 1\}$ in the nimber field is exactly the t-fold iterate of the Artin–Schreier map $\wp(y) = y^2 \oplus y$; this identifies $a(2^n + 2^{n-1} - 1)$ with $\wp^{n-1}(2^n)$. Second, this quantity obeys two recursions — one that strips the leading binary bit of n, one that handles n a power of 2 — which together reduce every case to $n = 1$.

1 The conjecture

Conway's nim-addition $\oplus$ is bitwise exclusive-or and nim-multiplication $\otimes$ is defined by the mex rule [1, Ch. 6]. OEIS `A059970` is the nim-factorial $a(1) = 1$, $a(n) = n \otimes a(n-1)$ for $n > 1$, contributed by John W. Layman on 5 March 2001. The entry carries the comment

> *Conjectures: (1) Nim-Factorial*$(2^n - 1) = 1$ *(verified for* $n = 1, 2, 3, \ldots, 16$*).*
> *(2) Nim-Factorial*$(2^n + 2^{n-1} - 1) = 2$ *(verified for* $n = 1, 2, 3, \ldots, 15$*).*

Part (1) is settled in a companion note; its result is restated as Theorem 2 below, with proof, so that the present note is self-contained. Here we settle part (2). As of the "Last modified" line on the live entry (23 August 2026) both statements are still recorded as unproven conjectures, and no proof appears in the entry's links, programs or in the literature on nimber arithmetic.

2 Notation and the first conjecture

Write $F_k = 2^{2^k}$ for the k-th Fermat 2-power and $K_k = \{0, 1, \ldots, F_k - 1\}$. We use three facts of Conway [1, Ch. 6]:

(N1) K_k is a field of order 2^{2^k} under $(\oplus, \otimes)$, and $K_0 \subset K_1 \subset \cdots$;

(N2) $F_k \otimes a = F_k \cdot a$ (ordinary product) for $a < F_k$;

(N3) $F_k \otimes F_k = F_k \oplus \frac{1}{2}F_k$.

Fix k and abbreviate $F = F_k$, $K = K_k$, $q = |K| = F$, and $\mathbb{F} = K_{k+1} \cong \mathrm{GF}(q^2)$. Powers are nim-powers.

Lemma 1. *$F^q = F \oplus 1$. Consequently $\bigotimes_{b\in K}(z \oplus b) = z^q \oplus z$ for every $z \in \mathbb{F}$, and $\bigotimes_{b\in K}\big((c \otimes F) \oplus b\big) = c$ for every $c \in K$.*

Proof. By (N3), F is a root of $X^2 \oplus X \oplus c_0$ with $c_0 = \frac{1}{2}F \in K$. As $F \notin K$ this is irreducible over K, so both roots lie in $\mathbb{F}$ and are swapped by $x \mapsto x^q$; they sum to 1, hence $F^q = F \oplus 1$. Next, $\prod_{b\in K}(X - b) = X^q - X$ in $K[X]$; evaluating at z in characteristic 2 gives the second claim. Finally for $z = c \otimes F$ with $c \in K$ we get $z^q = c^q \otimes F^q = c \otimes (F \oplus 1) = z \oplus c$, so $z^q \oplus z = c$. □

Theorem 2 (companion note)**.** *$\bigotimes_{v=1}^{2^n-1} v = 1$ for every $n \geq 0$. In particular $a(2^n-1) = 1$.*

Proof. Induction on n; the case $n = 0$ is the empty product. For $n \geq 1$ let 2^k be the largest power of 2 with $2^k \leq n$ and $m = n - 2^k$, so $0 \leq m < 2^k$ and $2^n = 2^m \cdot F$ with $F = F_k$. Every $v < 2^n$ is uniquely $a \cdot F + b$ with $a < 2^m$, $b \in K$; there are no carries, so $a \cdot F + b = (a \otimes F) \oplus b$ using (N2). Splitting the product on a, the term $a = 0$ contributes the product of all nonzero elements of the field K, which is $-1 = 1$ in characteristic 2, and each $a \neq 0$ contributes $\bigotimes_{b\in K}((a \otimes F) \oplus b) = a$ by Lemma 1. Hence $\bigotimes_{v=1}^{2^n-1} v = \bigotimes_{a=1}^{2^m-1} a$, which is 1 by induction. □

3 Subspace polynomials and the Artin–Schreier map

For $t \geq 0$ set $V_t = \{0, 1, \ldots, 2^t - 1\}$ and define, for a nimber x,

$$L_t(x) = \bigotimes_{j\in V_t}(x \oplus j).$$

V_t is an $\mathbb{F}_2$-subspace of the nimber field (it is closed under $\oplus$, with basis $1, 2, 4, \ldots, 2^{t-1}$), so L_t is the subspace polynomial of V_t and is therefore 2-linearized, hence additive: $L_t(x \oplus y) = L_t(x) \oplus L_t(y)$ [2, Thm. 3.52]. Let

$$\wp(y) = y^2 \oplus y$$

be the Artin–Schreier map.

Lemma 3. *$\bigotimes_{v=2^t}^{2^{t+1}-1} v = 1$ for every $t \geq 0$; equivalently $L_t(2^t) = 1$.*

Proof. All the factors are nonzero, so we may divide in the nimber field: $\bigotimes_{v=2^t}^{2^{t+1}-1} v = \big(\bigotimes_{v=1}^{2^{t+1}-1} v\big) \otimes \big(\bigotimes_{v=1}^{2^t-1} v\big)^{-1} = 1 \otimes 1^{-1} = 1$ by Theorem 2. Since $j < 2^t$ implies $2^t + j = 2^t \oplus j$, the left side is $L_t(2^t)$. □

Lemma 4. *$L_t(x) = \wp^t(x)$ for every $t \geq 0$ and every nimber x, where $\wp^t$ is the t-fold iterate.*

Proof. Induction on t. For $t = 0$, $L_0(x) = x$. For $t \geq 1$ write $V_t = V_{t-1} \cup (2^{t-1} \oplus V_{t-1})$, a disjoint union, so $L_t(x) = L_{t-1}(x) \otimes L_{t-1}(x \oplus 2^{t-1})$. By additivity of L_{t-1} and Lemma 3, $L_{t-1}(x \oplus 2^{t-1}) = L_{t-1}(x) \oplus 1$. Hence

$$L_t(x) = L_{t-1}(x) \otimes \big(L_{t-1}(x) \oplus 1\big) = L_{t-1}(x)^2 \oplus L_{t-1}(x) = \wp\big(L_{t-1}(x)\big) = \wp^t(x). \qquad \square$$

4 The second conjecture

For $n \geq 1$ put

$$g_n = \bigotimes_{v=2^n}^{2^n+2^{n-1}-1} v = \bigotimes_{j\in V_{n-1}} (2^n \oplus j) = L_{n-1}(2^n),$$

the middle equality because $j < 2^{n-1} < 2^n$ forces $2^n + j = 2^n \oplus j$. By Theorem 2,

$$a(2^n + 2^{n-1} - 1) = \Big(\bigotimes_{v=1}^{2^n-1} v\Big) \otimes g_n = g_n, \tag{1}$$

so conjecture (2) is exactly the assertion $g_n = 2$ for all $n \geq 1$.

Lemma 5 (leading-bit recursion)**.** *Let $n \geq 2$, let 2^k be the largest power of 2 with $2^k \leq n$, and suppose $m = n - 2^k \geq 1$. Then $g_n = g_m$.*

Proof. Keep $F = F_k$, $K = K_k$. Since $m < 2^k$ we have $m + 1 \leq 2^k$, so $2^{m+1} \leq 2^{2^k} = F$. Also $2^n = 2^m \cdot F = 2^m \otimes F$ by (N2), and $2^{n-1} = 2^{m-1} \cdot F$.

Every $j < 2^{n-1}$ is uniquely $a \cdot F + b$ with $a < 2^{m-1}$ and $b \in K$, and as before $a \cdot F + b = (a \otimes F) \oplus b$. Therefore

$$2^n \oplus j = (2^m \otimes F) \oplus (a \otimes F) \oplus b = \big((2^m \oplus a) \otimes F\big) \oplus b.$$

Put $c_a = 2^m \oplus a$. Since $a < 2^{m-1}$ we have $c_a < 2^{m+1} \leq F$, so $c_a \in K$ and Lemma 1 applies:

$$g_n = \bigotimes_{a\in V_{m-1}} \bigotimes_{b\in K} \big((c_a \otimes F) \oplus b\big) = \bigotimes_{a\in V_{m-1}} (2^m \oplus a) = g_m. \qquad \square$$

Lemma 6 (power-of-two case)**.** *$g_{2^k} = g_{2^k-1}$ for every $k \geq 1$.*

Proof. By (N3), $\wp(F_k) = F_k \otimes F_k \oplus F_k = (F_k \oplus \frac{1}{2}F_k) \oplus F_k = \frac{1}{2}F_k = 2^{2^k-1}$. Writing $n = 2^k$, so that $2^n = F_k$, Lemma 4 gives

$$g_{2^k} = L_{2^k-1}\big(F_k\big) = \wp^{2^k-1}(F_k) = \wp^{2^k-2}\big(\wp(F_k)\big) = \wp^{(2^k-1)-1}\big(2^{2^k-1}\big) = L_{2^k-2}\big(2^{2^k-1}\big) = g_{2^k-1},$$

where the last equality is the definition of g_n at $n = 2^k - 1 \geq 1$. $\square$

Theorem 7. *$g_n = 2$ for every $n \geq 1$; equivalently, $a(2^n + 2^{n-1} - 1) = 2$ for every $n \geq 1$. This is Layman's conjecture (2).*

Proof. Strong induction on n. For $n = 1$, $V_0 = \{0\}$ and $g_1 = 2 \oplus 0 = 2$.

Let $n \geq 2$ and let 2^k be the largest power of 2 with $2^k \leq n$. If $n \neq 2^k$ then $m = n - 2^k$ satisfies $1 \leq m < n$ and $g_n = g_m = 2$ by Lemma 5 and the induction hypothesis. If $n = 2^k$ then $k \geq 1$ and $g_n = g_{n-1} = 2$ by Lemma 6 and the induction hypothesis. The equivalence with the statement about a is (1). $\square$

Remark. Two structural facts fall out on the way and may be of independent interest. First, Lemma 4: the subspace polynomial of $\{0, \dots, 2^t - 1\}$ in the nimber field is the t-fold iterate of $y \mapsto y^2 \oplus y$, so that $a(2^n + 2^{n-1} - 1) = \wp^{n-1}(2^n)$. Second, combining Lemma 3 with Lemma 4 shows $\wp^t(2^t) = 1$ for all t. Neither is deep — everything here is Conway's field structure plus the Artin–Schreier map — and the natural destination is an OEIS comment rather than a journal.

5 Verification

A verification script, written separately from this note, checks every assertion made above and prints every number quoted. It (i) implements nim-multiplication twice by genuinely different algorithms, Conway's recursion on Fermat 2-powers and a direct tabulation from the mex definition, and confirms they agree on $\{0,\dots,31\}^2$ with 0 mismatches; (ii) reproduces the first 70 published terms of `A059970` with 0 mismatches; (iii) confirms $F \otimes F = F \oplus \frac{1}{2}F$ and $\wp(F) = \frac{1}{2}F$ for $F = 2, 4, 16, 256, 65536$, obtaining $\wp(F) = 1, 2, 8, 128, 32768$ respectively, and $F^q = F \oplus 1$ for $k \le 3$; (iv) confirms Lemma 1 in 148 and 22 instances respectively; (v) confirms Lemma 3 for $t = 0,\dots,9$ and Lemma 4 in 126 instances with $t \le 8$; (vi) confirms $g_n = 2$ for $n = 1,\dots,14$ together with Lemma 5 in all 10 applicable cases and Lemma 6 for $k = 1, 2, 3$; and (vii) computes $a(2^n + 2^{n-1} - 1)$ from the defining recursion $a(n) = n \otimes a(n-1)$ for $n = 1,\dots,17$, obtaining 2 in every case. The largest index reached was $a(196607) = 2$; the conjecture as posted was verified only to $n = 15$.

References

[1] J. H. Conway, *On Numbers and Games*, 2nd ed., A K Peters, 2001, Chapter 6.

[2] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997, Theorem 3.52.

[3] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A059970`, entry `A059970`, consulted 23 August 2026.

A Suggested OEIS comment (plain text, house style)

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Conjecture (2) is true. Since a(2^n-1) = 1 (see the comment on conjecture
(1)), the claim reduces to g(n) = 1 nim-times the product of the nimbers
2^n, 2^n+1, ..., 2^n+2^(n-1)-1, i.e. to g(n) = 2. Two facts. (a) The
subspace polynomial L_t(x) = prod_{j < 2^t} (x XOR j) satisfies
L_t(x) = P^t(x), where P(y) = y^2 XOR y: indeed L_t(x) = L_{t-1}(x) *
(L_{t-1}(x) XOR L_{t-1}(2^(t-1))) and L_{t-1}(2^(t-1)) = 1 because the
nim-product of 2^(t-1), ..., 2^t-1 is 1. Hence g(n) = P^(n-1)(2^n).
(b) Two recursions. If 2^k is the largest power of 2 at most n and
m = n - 2^k >= 1, then writing F = 2^(2^k) and splitting each j < 2^(n-1)
as (a*F) XOR b with a < 2^(m-1), b < F, and using
prod_{b < F} ((c*F) XOR b) = c, one gets g(n) = g(m). If n = 2^k, then
P(F_k) = F_k/2 by F_k*F_k = 3F_k/2, so g(2^k) = P^(2^k-1)(F_k) =
P^(2^k-2)(2^(2^k-1)) = g(2^k-1). Both recursions decrease n, and
g(1) = 2, so g(n) = 2 for all n >= 1.
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